Propositional Logic



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Propositions

- A proposition is a declarative sentence that is either true or false, but not both.

- The following are propositions
  - The Moon is made of cheese.
  - Toronto is the capital of Canada.
  - $1 + 0 = 1$
  - 4 is a prime

- The following are not propositions
  - What time is it?
  - Buy four books
  - $x + y = 3$
  - $x + 1 = 2$



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Propositions

- Propositions are usually represented by variables called propositional variables.
  - The conventional letters for propositional variables are $p, q, r, s, \ldots$

- Propostions take on truth values that is either True (denoted by T or 1) or False (denoted by F or 0)

- The area of logic that deals with propositions is called propositional logic or propositional calculus.



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Compound Propositions

- Construction of propositions
  - New propositions, called compound propositions, are formed from existing propositions using logical operators.
  - Truth value of a compound proposition depends on the truth value of its atomic propositions.

- The logical operators are
  - negation $\neg$
  - conjunction $\wedge$
  - disjunction $\vee$
  - exclusive or $\oplus$
  - implication $\rightarrow$
  - biconditional $\leftrightarrow$



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Negation

- Let p be a proposition. The negation of p is a proposition *not* p. It is denoted by $\neg p$.
  - If p is true, $\neg p$ is false
  - If p is false, $\neg p$ is true
  - $\neg p$ is not actually connective, i.e., it doesnt join two of anything
  - *not* is a unary operation on propostions
- The truth table for the negation of a proposition.

p	$\neg p$
T	F
F	T

or

p	$\neg p$
1	0
0	1

- **Example**: If p : The earth is round, then $\neg p$: It is not the case that the earth is round, or more simply, The earth is not round.



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Conjunction

- Let p and q be propositions. The *conjunction* of p and q is the proposition "p and q".
  - Denoted by $p \wedge q$
  - The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

- The truth table for the conjunction of two propositions.

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1



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Examples of conjunction

- Example 1.
  - p : I am at home.
  - q : It is raining.
  - $p \wedge q$: I am at home and it is raining.
- Example 2.
  - p : Rebecca's PC has more than 16 GB free hard disk space.
  - q : The processor in Rebecca's PC runs faster than 1 GHz.
  - $p \wedge q$: Rebeccas PC has more than 16 GB free hard disk space, and its processor runs faster than 1 GHz.



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Disjunction

- Let p and q be propositions. The *disjunction* of p and q is the proposition "p or q".
  - Denoted by $p \vee q$
  - The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

- The truth table for the disjunction of two propositions.

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1



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Examples of disjunction

- Example 1.
  - p : I am at home.
  - q : It is raining.
  - $p \vee q$: I am at home or it is raining.
- Example 2.
  - p : Rebecca's PC has more than 16 GB free hard disk space.
  - q : The processor in Rebecca's PC runs faster than 1 GHz.
  - $p \vee q$: Rebeccas PC has more than 16 GB free hard disk space, or its processor runs faster than 1 GHz.



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Exclusive disjunction (Exclusive or)

- Let p and q be propositions. The *exclusive or* of p and q is the proposition "either p or q, but not both".
  - Denoted by $p \oplus q$
  - The exclusive or $p \oplus q$ is true when exactly one of p and q is true and is false otherwise.

- The truth table for the exclusive or of two propositions.

p	q	$p \oplus q$
0	0	0
0	1	1
1	0	1
1	1	0



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Exclusive versus Inclusive

- Depending on the circumstances, some disjunctions are inclusive and some are exclusive.
- Examples of Inclusive
  - "I have a dog" or "I have a cat"
  - "It is warm outside" or "It is raining"
- Examples of Exclusive
  - Today is either Tuesday or it is Thursday
  - Pat is either male or female



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Conditional statement (Implication)

- Let p and q be propositions. The conditional statement $p \to q$ is the proposition "if p, then q."
  - In $p \to q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).
  - $p \to q$ is false when p is true and q is false, and true otherwise.

- The truth table for the implication.

p	q	$p \to q$
0	0	1
0	1	1
1	0	0
1	1	1



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Different Ways of Expressing $p \to q$

"if p, then q"
"if p, q"
"p is sufficient for q"
"q if p"
"q when p"
"q unless $\neq p$"
"a necessary condition for p is q"
"p implies q"
"p only if q"
"a sufficient condition for q is p"
"q whenever p"
"q is necessary for p"
"q follows from p"

- In $p \to q$ there does not need to be any connection between the antecedent or the consequent. The "meaning" of $p \to q$ depends only on the truth values of p and q.



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Examples of implication

- Example 1.
  - p : 1+1 = 3.
  - q : Your grandma wears combat boots.
  - $p \to q$: If 1+1=3, then your grandma wears combat boots.
- Example 2.
  - p : Maria learns discrete mathematics.
  - q : Maria will find a good job.
  - $p \to q$: If Maria learns discrete mathematics, then she will find a good job.



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Converse, Contrapositive, and Inverse

- We can form some new conditional statements starting with a conditional statement $p \to q$.
  - The proposition $q \to p$ is called the **converse** of $p \to q$
  - The proposition $\neg q \to \neg p$ is called the **contrapositive** of $p \to q$
  - The proposition $\neg p \to \neg q$ is called the **inverse** of $p \to q$

Example: Consider the proposition "The home team wins whenever it is raining?"

- The original statement can be rewritten as "If it is raining, then the home team wins."
- The converse is "If the home team wins, then it is raining."
- The inverse is "If it is not raining, then the home team does not win."
- The contrapositve is "If the home team does not win, then it is not raining."



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Truth table of converse, contrapositive, and inverse

p	q	$\neg p$	$\neg q$	$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
0	0	1	1	1	1	1	1
0	1	1	0	1	0	0	1
1	0	0	1	0	1	1	0
1	1	0	0	1	1	1	1

- From the above truth table it is evident that the contrapositive, $\neg q \to \neg p$, of a conditional statement $p \to q$ always has the same truth value as $p \to q$.



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Biconditional

- Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition "p if and only if q."
  - The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.
  - Biconditional statements are also called bi-implications.
  - Bi-implication $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

- The Truth Table for the Biconditional $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
0	0	1
0	1	0
1	0	0
1	1	1



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Examples of bi-implication

- Example:
  - p : Maria learns discrete mathematics.
  - q : Maria will find a good job.
  - $p \leftrightarrow q$: Maria learns discrete mathematics if and only if she will find a good job.

- Some other common ways to express $p \leftrightarrow q$ are
  - "p is necessary and sufficient for q"
  - "if p then q, and conversely"
  - "p iff q"



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Truth table of a compound proposition

- The truth table of a compound proposition with n atomic propositions has rows and columns.
  - The first n colunms are for n variables.
  - There are 2^n possible assignments for the variables.
  - There are 2^n rows.
  - Rows represent the binary numbers from 0 to $2^n - 1$.
- Two propostions are said to be logically equivalent if they have same truth values.
  - If p and q are logically equivalent, then we write $p \Leftrightarrow q$ or $p \equiv q$.
  - Example: $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$



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Precedence of logical operators

- We use parentheses to specify the order in which logical operators in a compound proposition are to be applied.
- In order to reduce the number of parentheses, we use the following precedence of logical operators.

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

- With this precedence $p \wedge q \vee r$ means $(p \wedge q) \vee r$ rather than $p \wedge (q \vee r)$.



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Exercise

a) Construct a truth table for each of these compound propositions

1. $(p \vee \neg q) \rightarrow q$
2. $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
3. $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
4. $(p \oplus q) \rightarrow (p \wedge q)$

b) Let p and q be the propostions

p : You drive over 65 miles per hour.

q : You get a spending ticket.

Write these propositions using p and q and logical connectives (including negations).

1. You will get a speeding ticket if you drive over 65 miles per hour.
2. Driving over 65 miles per hour is sufficient for getting a speeding ticket.



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3. You do not drive over 65 miles per hour.
4. You drive over 65 miles per hour, but you do not get a speeding ticket.
5. If you do not drive over 65 miles per hour, then you will not get a speeding ticket.
6. You get a speeding ticket, but you do not drive over 65 miles per hour.
7. Whenever you get a speeding ticket, you are driving over 65 miles per hour.



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